Business Insights Assessment

Project Overview

- **Objective:** We need a unified data view that helps us understand customer behavior, spending patterns, and overall business performance across all restaurant locations and platforms.

  The system should calculate and update Customer Lifetime Value (CLV) daily, classify customers into high, medium, and low value groups, and segment them using recency, frequency, and monetary (RFM) metrics. It should identify customers at risk of churn, highlight sales trends and seasonal patterns, and compare the performance of loyalty members versus non-members.

  Additionally, we need insights into the best- and worst-performing locations and the impact of discounts or promotions on revenue and profitability. All these insights should be accessible through an interactive dashboard to support data-driven marketing, operational, and strategic decisions.

Step-by-Step Plan

Step 1: Source Data Explanation

1. **Task:**
   - Download the orders related data files from Google Drive.
   - Verify the integrity of the downloaded files (e.g., check for missing or corrupt values).

   All the files are present in the google drive link below
   order_items **order_item_options** **date_dim**

1. order_items *(203,519 records)*

Captures transactional-level data for each item in a customer's order.

Column Name	Data Type	Description
app_name	String	Name of the ordering platform or channel used
restaurant_id	String / Int	Unique identifier for the restaurant
creation_time_utc	Timestamp	Timestamp of when the order was placed (UTC)
order_id	String / Int	Unique identifier for the order
user_id	String / Int	Unique identifier for the customer placing the order
printed_card_number	String	Tokenized or masked loyalty card number
is_loyalty	Boolean	Flag indicating loyalty membership (True/False)
currency	String	Currency of the transaction (e.g., USD)
lineitem_id	String / Int	Unique identifier for the specific item in the order
item_category	String	Category of the item (e.g., Beverage, Entree, etc.)
item_name	String	Name of the item
item_price	Decimal	Unit price of the item

Column Name	Data Type	Description
item_quantity	Integer	Quantity of the item purchased

2. order_item_options *(193,017 records)*

Details add-ons, customizations, or modifiers associated with order items.

Column Name	Data Type	Description
order_id	String / Int	Identifier linking to the parent order
lineitem_id	String / Int	Identifier linking to the specific order item
option_group_name	String	Group category for options (e.g., Size, Toppings)
option_name	String	Selected option or customization (e.g., Extra Cheese)
option_price	Decimal	Price of the option (negative if it's a discount)
option_quantity	Integer	Number of times the option was added

3. date_dim *(Date Dimension Table)*

Used for time-based joins and calendar aggregations.

Column Name	Data Type	Description
date_key	Date	Full calendar date
day_of_week	String	Day of the week (e.g., Monday)
week	Integer	Week number in the year
month	String	Month name (e.g., January)
year	Integer	Calendar year
is_weekend	Boolean	Whether the date falls on a weekend
is_holiday	Boolean	Flag indicating if the date is a recognized holiday
holiday_name	String	Name of the Holiday

Step 2: Initial Data Analysis

1. **Task:**
    - Analyze the CSV files to understand the schema, relationships, and data quality issues.(Manually)
    - Identify key columns for joins and insights.(Manually)

Step 3: Define Data Engineering Pipeline/Architecture

1. **Task:**
   - Design a pipeline architecture for ingesting, transforming, and storing data. Design a production level pipeline that includes scheduling, encryption, failure reload mechanism.
   - Design an architecture based on below requirements:

     Consider your source as SQL Server i.e you need to pull data from your database tables. A video step to load data in SQL Server can be found here. (WINDOWS OS)

     The entire architecture should be designed using *AWS resources*. The company does not want to get new licenses. The company does not want to use Snowflake, DBT, etc external tools.

     All logics should be done using *PYSPARK.*

     Please schedule the pipeline jobs once daily as a batch process.

     Define a Data Model that satisfies below requirement:

     - **Customer Lifetime Value (LTV), and how would you structure it to show how LTV evolves for each customer daily? (Primary Goal)**
     - **Look into Step 5 for additional business requirements while creating your data model. (Secondary Goal)**

     Present the architecture for Subject Matter Expert (SME) approval.

     Move head once clearance from the SME

Deliverable:

- Pipeline architecture diagram and explanation.
- Create a solution design document.
- Written approval from the SME.
- Use tools like draw.io, miro, etc for diagrams.

Step 4: Build the Data Pipeline

Deliverable:

- Working pipeline that can handle raw data ingestion, transformation in AWS.

Step 5: Define and Calculate Metrics as per Business Requirement:

Primary Metrics:

1. **Customer Lifetime Value (CLV):**

Goal: Estimate how much total revenue a customer will generate over their entire relationship with the business.

Why it matters: Helps prioritize high-value customers, plan marketing budgets wisely, and improve retention strategies.

How to do it:

- Use order_items and order_item_options to compute revenue per order.
- Aggregate total spend per customer_id**.**

Group CLV values (for tagging):

- High CLV: Top 20% customers
- Medium CLV: Mid 60%
- Low CLV: Bottom 20%

Secondary Metrics

2. **Customer Segmentation & Behavior:**

Goal: Group customers based on spending and activity to support campaign targeting.

Why it matters: Enables personalized offers and engagement.

How to do it:

- **Use RFM logic based on order_items:**
    - Recency: Days since last purchase
    - Frequency: Number of purchases in last N months
    - Monetary: Total spend in last N months
- **Segment:**
    - VIPs: High R, F, M
    - New Customers: Low F, high R
    - Churn Risk: Low R, low F

3. **Churn Indicators:**

Goal: Build a customer activity profile to help marketing identify at-risk customers (no predictions).

Why it matters: Allows timely retention actions by analysts.

How to do it:

- For each customer_id, compute:

  • Days since last order
  • Average gap between orders
  • % change in spend over last N periods

- Tag customers based on inactivity thresholds (e.g., >45 days = “at risk”)

4. **Sales Trends Monitoring:**

Goal: Generate time-based summaries to analyze sales patterns.

Why it matters: Helps identify peak periods and plan resources.

How to do it:

  - Aggregate daily, weekly, and monthly revenue from order_items
  - Break down by:
    • Location
    • Menu category (if available)
    • Time of day (optional)

5. **Loyalty Program Impact**

Goal: Compare loyalty members vs non-members in terms of spend and engagement.

Why it matters: Evaluates ROI of the loyalty program.

How to do it:

  - Filter order_items by is_loyalty = true vs false
  - Compare per-customer:
    • Average Spend
    • Repeat Orders
    • Lifetime Value

6. **Top-Performing Locations**

Goal: Identify best and worst-performing store locations.

Why it matters: Informs decisions about promotions, staffing, or expansion.

How to do it:

- Group order_items by location_id (or store_id if available)
- Calculate:
  • Total revenue

- Average order value
- Orders per day/week
- Rank locations based on revenue

7. **Pricing & Discount Effectiveness**

Goal: Measure how discounts affect revenue and profitability.

Why it matters: Helps optimize pricing strategies.

How to do it:

- Use order_item_options.option_price to detect discounts (option_price < 0)
- Compare:
  - Revenue from discounted orders vs non-discounted
  - Number of orders before/after applying discounts

Step 6: Build the Dashboard using Streamlit:

Please go through this link for working on Streamlit for Visualization:

Streamlit Tutorial

1. **Task:**

   Design an interactive dashboard showcasing the calculated metrics and insights.

   Deliverable:

   - **Customer Segmentation Dashboard:**
     - **Question:** What distinct customer segments emerge when grouping customers by purchase behavior (total spend, frequency, recency) and loyalty status?
     - **Focus:** Visualize segmentation (e.g., via RFM scores) to enable targeted marketing.
   - **Churn Risk Indicators Dashboard:**
     - **Question:** Which metrics—such as days since last order, average order interval, and spend trends—correlate with a higher churn risk?
     - **Focus:** Highlight threshold-based alerts (e.g., customers at risk) to prompt re-engagement actions.

- **Sales Trends and Seasonality Dashboard:**
    - **Question:** What are the monthly and seasonal trends in sales, and how do these vary by product category or location?
    - **Focus:** Track weekly/monthly sales aggregates and holiday spikes to support inventory planning and staffing decisions.
- **Loyalty Program Impact Dashboard:**
    - **Question:** How does loyalty membership affect customer spending and repeat order rates?
    - **Focus:** Compare key metrics (CLV, average order value, repeat purchase rate) between loyalty and non-loyalty customers to assess program effectiveness and inform potential adjustments.
- **Location Performance Dashboard:**
    - **Question:** Which restaurant locations generate the highest revenue, and what operational metrics (e.g., average order size, customer retention) distinguish top performers?
    - **Focus:** Rank locations and highlight actionable insights for expansion or targeted improvements.
- **Pricing and Discount Effectiveness Dashboard:**
    - **Question:** How are discounts and promotions affecting overall sales volume and net revenue?
    - **Focus:** Compare revenue and profit (gross versus net after discount adjustments) for discounted versus full-price transactions to refine pricing strategies.

Step 7: Submission and Review

1. **Task:**
    - Submit all deliverables, including:
        - Pipeline documentation.
        - Code files (ETL scripts, Pyspark SQL queries).
        - Setup Configurations (if any)
        - Final dashboard.
        - Create a CI/CD pipeline using github.
2. **Instructions:**
    - Ensure all files are properly named and documented.
    - **Include a short video or presentation explaining the work**.
    - Explain **'WHY**' behind each tech stack used in your design.

Thank You.